

Homework 1

SI252 Reinforcement Learning

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Problem 1. Random variables

A random variable lets us focus on a numerical feature of a random experiment rather than its full outcome. For example, when tossing two coins, the number of heads treats HT and TH alike, ignoring the order when it is irrelevant. This makes the quantity of interest easier to study mathematically.

Although formally a function $X : \Omega \rightarrow \mathbb{R}$, it is called a “random variable” because it represents a quantity whose value depends on the random outcome. The assignment rule is fixed; what is uncertain is which outcome occurs and hence which value the quantity takes.

Problem 2. Probability and estimation identities

(a)

1. Both X and Y are discrete

By the definition of conditional probability,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}.$$

Also,

$$P(X = x, Y = y) = P(X = x \mid Y = y)P(Y = y).$$

Substituting gives

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}.$$

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. For every measurable set $B \subseteq \mathbb{R}$, conditioning on Y yields

$$P(X = x, Y \in B) = \int_B P(X = x \mid Y = y) f_Y(y) dy.$$

Hence,

$$\begin{aligned} P(Y \in B \mid X = x) &= \frac{P(X = x, Y \in B)}{P(X = x)} \\ &= \int_B \frac{P(X = x \mid Y = y)f_Y(y)}{P(X = x)} dy. \end{aligned}$$

By the definition of a conditional density, the integrand is a density of Y conditional on $X = x$. Therefore,

$$f_{Y|X}(y \mid x) = \frac{P(X = x \mid Y = y)f_Y(y)}{P(X = x)}.$$

3. X is continuous and Y is discrete

Fix y such that $P(Y = y) > 0$. For every measurable set $A \subseteq \mathbb{R}$,

$$\begin{aligned} P(X \in A, Y = y) &= P(Y = y)P(X \in A \mid Y = y) \\ &= \int_A f_{X|Y}(x \mid y)P(Y = y) dx. \end{aligned}$$

On the other hand, conditioning on X gives

$$P(X \in A, Y = y) = \int_A P(Y = y \mid X = x)f_X(x) dx.$$

Since these integrals are equal for every measurable set A , their integrands are equal almost everywhere:

$$P(Y = y \mid X = x)f_X(x) = f_{X|Y}(x \mid y)P(Y = y).$$

Dividing by $f_X(x) > 0$, we obtain

$$P(Y = y \mid X = x) = \frac{f_{X|Y}(x \mid y)P(Y = y)}{f_X(x)}.$$

4. Both X and Y are continuous and admit a joint density

Let $f_{X,Y}(x, y)$ denote the joint density. By the definition of conditional density,

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Moreover,

$$f_{X,Y}(x, y) = f_{X|Y}(x \mid y)f_Y(y).$$

Substituting yields

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y)f_Y(y)}{f_X(x)}.$$

(b)

1. Both X and Y are discrete

The events $\{Y = y\}$, taken over the possible values of Y , form a countable partition of the sample space up to a null set. Therefore, by countable additivity,

$$P(X = x) = \sum_y P(X = x, Y = y).$$

Using the definition of conditional probability,

$$P(X = x, Y = y) = P(X = x | Y = y)P(Y = y).$$

Consequently,

$$P(X = x) = \sum_y P(X = x | Y = y)P(Y = y).$$

The sum may be restricted to values y for which $P(Y = y) > 0$.

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. By the previously established Bayes' rule,

$$f_{Y|X}(y | x) = \frac{P(X = x | Y = y)f_Y(y)}{P(X = x)}.$$

Rearranging gives

$$P(X = x | Y = y)f_Y(y) = P(X = x)f_{Y|X}(y | x).$$

Integrating both sides with respect to y , we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} P(X = x | Y = y)f_Y(y) dy &= P(X = x) \int_{-\infty}^{\infty} f_{Y|X}(y | x) dy \\ &= P(X = x). \end{aligned}$$

3. X is continuous and Y is discrete

Let $A \subseteq \mathbb{R}$ be any measurable set. Applying the discrete law of total probability to the event $\{X \in A\}$, we have

$$P(X \in A) = \sum_y P(X \in A | Y = y)P(Y = y).$$

By the definition of conditional density,

$$P(X \in A | Y = y) = \int_A f_{X|Y}(x | y) dx.$$

Thus,

$$\begin{aligned} P(X \in A) &= \sum_y \left[\int_A f_{X|Y}(x | y) dx \right] P(Y = y) \\ &= \int_A \left[\sum_y f_{X|Y}(x | y)P(Y = y) \right] dx. \end{aligned}$$

The interchange of the sum and the integral is justified by Tonelli's theorem, since all the terms are nonnegative.

On the other hand,

$$P(X \in A) = \int_A f_X(x) dx.$$

Since these expressions agree for every measurable set A , uniqueness of densities yields

$$f_X(x) = \sum_y f_{X|Y}(x | y)P(Y = y).$$

4. Both X and Y are continuous and admit a joint density

Assume that X and Y have a joint density $f_{X,Y}(x, y)$. By marginalization,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

The conditional density satisfies

$$f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y)$$

almost everywhere. Substituting gives

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x | y)f_Y(y) dy.$$

(c)

$$\text{Var}(Y | X) = E[Y^2 | X] - (E[Y | X])^2.$$

Taking expectations and applying the law of total expectation gives

$$\begin{aligned} E[\text{Var}(Y | X)] &= E[E[Y^2 | X]] - E[(E[Y | X])^2] \\ &= E[Y^2] - E[(E[Y | X])^2]. \end{aligned}$$

Using the variance identity $\text{Var}(Z) = E[Z^2] - (E[Z])^2$, we obtain

$$\begin{aligned} \text{Var}(E[Y | X]) &= E[(E[Y | X])^2] - (E[E[Y | X]])^2 \\ &= E[(E[Y | X])^2] - (E[Y])^2, \end{aligned}$$

where the last equality follows from the law of total expectation.

Adding the expressions above, the middle terms cancel:

$$\begin{aligned} &E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]) \\ &= E[Y^2] - E[(E[Y | X])^2] + E[(E[Y | X])^2] - (E[Y])^2 \\ &= E[Y^2] - (E[Y])^2 \\ &= \text{Var}(Y). \end{aligned}$$

Therefore,

$$\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]).$$

(d)

Assume that $E[X^2] < \infty$, $E[Y^2] < \infty$, and $\text{Var}(X) > 0$. We seek constants a and b that minimize

$$Q(a, b) = E[(Y - a - bX)^2].$$

$$Q(a, b) = E[Y^2] - 2aE[Y] - 2bE[XY] + a^2 + 2abE[X] + b^2E[X^2].$$

$$\frac{\partial Q}{\partial a} = -2E[Y] + 2a + 2bE[X] = 0.$$

Hence,

$$a = E[Y] - bE[X].$$

Similarly,

$$\frac{\partial Q}{\partial b} = -2E[XY] + 2aE[X] + 2bE[X^2] = 0,$$

so

$$E[XY] = aE[X] + bE[X^2].$$

Substituting $a = E[Y] - bE[X]$, we obtain

$$E[XY] = E[X]E[Y] - b(E[X])^2 + bE[X^2].$$

Rearranging,

$$E[XY] - E[X]E[Y] = b(E[X^2] - (E[X])^2).$$

By the definitions of covariance and variance,

$$\text{Cov}(X, Y) = b \text{Var}(X).$$

Therefore,

$$b = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Thus,

$$L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

The objective is a convex quadratic in (a, b) , with a positive-definite Hessian when $\text{Var}(X) > 0$, so this stationary point is the unique minimum. If $\text{Var}(X) = 0$, X is constant almost surely and the best predicted value is simply $E[Y]$.

(e)

Let $m(X) = \mathbb{E}[Y | X]$, and assume that Y has a finite second moment. For every measurable ϕ with $\mathbb{E}[\phi(X)^2] < \infty$, the tower property gives

$$\begin{aligned} \mathbb{E}[(Y - m(X))\phi(X)] &= \mathbb{E}[\mathbb{E}[(Y - m(X))\phi(X) | X]] \\ &= \mathbb{E}[\phi(X)(\mathbb{E}[Y | X] - m(X))] \\ &= 0. \end{aligned}$$

This proves the orthogonality property.

Conversely, assume that $g(X)$ has a finite second moment and that, for every such measurable function ϕ ,

$$\mathbb{E}[(Y - g(X))\phi(X)] = 0.$$

By the orthogonality property just proved,

$$\mathbb{E}[(Y - m(X))\phi(X)] = 0.$$

Subtracting the second equality from the first gives

$$\mathbb{E}[(m(X) - g(X))\phi(X)] = 0.$$

Since this holds for every such ϕ , we may choose

$$\phi(X) = m(X) - g(X),$$

which is square-integrable. Therefore,

$$\mathbb{E}[(m(X) - g(X))^2] = 0.$$

Since a nonnegative random variable with zero expectation is zero almost surely, $g(X) = m(X) = \mathbb{E}[Y | X]$ almost surely.

(f)

Proof. Let $\mu = \mathbb{E}[Z]$. By Chebyshev's inequality,

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

We consider two cases:

- If $a^2 \leq \sigma^2$, then

$$\mathbb{P}(|Z - \mu| \geq a) \leq 1 \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

- If $a^2 > \sigma^2$, then $\sigma^2 + a^2 \leq 2a^2$, and hence

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2} \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

Therefore, for every $a > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq a) \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

(g)

Proof. Let

$$Y_k = X_k - \mathbb{E}[X_k], \quad V = \sum_{k=1}^n (b_k - a_k)^2.$$

Then the Y_k are independent, $\mathbb{E}[Y_k] = 0$, and

$$a_k - \mathbb{E}[X_k] \leq Y_k \leq b_k - \mathbb{E}[X_k].$$

Thus, by Hoeffding's lemma, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda Y_k}] \leq \exp\left(\frac{\lambda^2 (b_k - a_k)^2}{8}\right).$$

Using independence, we obtain

$$\begin{aligned} \mathbb{E}[e^{\lambda(S_n - \mu)}] &= \prod_{k=1}^n \mathbb{E}[e^{\lambda Y_k}] \\ &\leq \exp\left(\frac{\lambda^2 V}{8}\right). \end{aligned}$$

Assume $V > 0$ and $t > 0$. By Markov's inequality, for every $\lambda > 0$,

$$\begin{aligned} \mathbb{P}(S_n - \mu \geq t) &= \mathbb{P}(e^{\lambda(S_n - \mu)} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda(S_n - \mu)}] \\ &\leq \exp\left(-\lambda t + \frac{\lambda^2 V}{8}\right). \end{aligned}$$

Choosing $\lambda = 4t/V$ gives

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Applying the same argument to the independent random variables $-X_1, \dots, -X_n$, whose interval lengths remain $b_k - a_k$, yields

$$\mathbb{P}(S_n - \mu \leq -t) = \mathbb{P}(\mu - S_n \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Therefore, by the union bound,

$$\begin{aligned} \mathbb{P}(|S_n - \mu| \geq t) &\leq \mathbb{P}(S_n - \mu \geq t) + \mathbb{P}(S_n - \mu \leq -t) \\ &\leq 2 \exp\left(-\frac{2t^2}{V}\right). \end{aligned}$$

For $t = 0$, the inequality holds trivially since $1 \leq 2$. Hence, when $V > 0$,

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\right), \quad t \geq 0.$$

If $V = 0$, all X_k are constant almost surely, so $S_n = \mu$ almost surely. In this degenerate case, the tail probability is 0 for $t > 0$ and 1 for $t = 0$.

Problem 3. Exponential conditional expectations

Assume that $X_i \sim \text{Exp}(\lambda_i)$, where λ_i is the rate parameter. Then

$$f_{X_i}(x) = \lambda_i e^{-\lambda_i x}, \quad x > 0.$$

By the memoryless property of the exponential distribution,

$$X_i - t \mid X_i > t \sim \text{Exp}(\lambda_i).$$

Therefore,

$$E(X_1 \mid X_1 > 2026) = 2026 + E(X_1) = \boxed{2026 + \frac{1}{\lambda_1}}.$$

For the second term,

$$E(X_1 \mid X_1 < 2026) = \frac{\int_0^{2026} x \lambda_1 e^{-\lambda_1 x} dx}{P(X_1 < 2026)}.$$

Evaluating the integral gives

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026 e^{-2026 \lambda_1}}{1 - e^{-2026 \lambda_1}}$$

or, equivalently,

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026}{e^{2026 \lambda_1} - 1}$$

Finally, since X_1, X_2, X_3 are independent, we can apply the memoryless property to each variable:

$$\begin{aligned} E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) \\ = \left(1997 + \frac{1}{\lambda_1}\right) + \left(2014 + \frac{1}{\lambda_2}\right) + \left(2026 + \frac{1}{\lambda_3}\right). \end{aligned}$$

Hence,

$$E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) = 6037 + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$$

Problem 4. Bivariate normal construction

Let Z_1, Z_2 be independent standard normal random variables, and define

$$X = \sigma_X Z_1 + \mu_X, \quad Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y,$$

where $\sigma_X, \sigma_Y > 0$ and $-1 < \rho < 1$.

(a) For any $a, b \in \mathbb{R}$,

$$aX + bY = a\mu_X + b\mu_Y + (a\sigma_X + b\rho\sigma_Y)Z_1 + b\sigma_Y\sqrt{1 - \rho^2}Z_2.$$

Since a linear combination of independent normal random variables is normal, $aX + bY$ is normal. Therefore, X and Y are jointly normal.

(b) We have

$$\text{Var}(X) = \sigma_X^2, \quad \text{Var}(Y) = \sigma_Y^2(\rho^2 + 1 - \rho^2) = \sigma_Y^2.$$

Using independence,

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_X \sigma_Y \left[\rho \text{Var}(Z_1) + \sqrt{1 - \rho^2} \text{Cov}(Z_1, Z_2) \right] \\ &= \rho \sigma_X \sigma_Y. \end{aligned}$$

Hence,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho$$

(c) The inverse transformation is

$$z_1 = \frac{x - \mu_X}{\sigma_X}, \quad z_2 = \frac{\frac{y - \mu_Y}{\sigma_Y} - \rho \frac{x - \mu_X}{\sigma_X}}{\sqrt{1 - \rho^2}}.$$

Its Jacobian determinant has absolute value

$$\left| \frac{\partial(z_1, z_2)}{\partial(x, y)} \right| = \frac{1}{\sigma_X \sigma_Y \sqrt{1 - \rho^2}}.$$

Since

$$f_{Z_1, Z_2}(z_1, z_2) = \frac{1}{2\pi} \exp\left(-\frac{z_1^2 + z_2^2}{2}\right),$$

the change-of-variables formula gives

$$\begin{aligned} f_{X, Y}(x, y) &= \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \\ &\times \exp\left\{-\frac{1}{2(1 - \rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right]\right\}, \end{aligned}$$

for all $x, y \in \mathbb{R}$.

Problem 5. A three-state Markov chain

The transition matrix, with states ordered as 1, 2, 3, is

$$P = \begin{pmatrix} \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix},$$

where

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

(a)

Since the chain is time-homogeneous, the one-step transition probabilities do not depend on time. Therefore,

$$P(X_5 = 3 \mid X_4 = 2) = p_{23} = \frac{1}{2},$$

and

$$P(X_4 = 1 \mid X_3 = 2) = p_{21} = \frac{1}{2}.$$

(b)

Let $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3)$ be the stationary distribution. It satisfies

$$\boldsymbol{\pi}P = \boldsymbol{\pi}, \quad \pi_1 + \pi_2 + \pi_3 = 1.$$

Thus,

$$\begin{cases} \pi_1 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{2}\pi_3, \\ \pi_2 = \frac{1}{4}\pi_3, \\ \pi_3 = \frac{3}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{4}\pi_3. \end{cases}$$

Substituting $\pi_2 = \frac{1}{4}\pi_3$ into the first equation gives

$$\frac{3}{4}\pi_1 = \frac{1}{8}\pi_3 + \frac{1}{2}\pi_3 = \frac{5}{8}\pi_3,$$

so

$$\pi_1 = \frac{5}{6}\pi_3.$$

Using the normalization condition,

$$\left(\frac{5}{6} + \frac{1}{4} + 1\right)\pi_3 = 1,$$

which yields

$$\pi_3 = \frac{12}{25}, \quad \pi_2 = \frac{3}{25}, \quad \pi_1 = \frac{2}{5}.$$

Hence,

$$\boldsymbol{\pi} = \left(\frac{2}{5}, \frac{3}{25}, \frac{12}{25}\right).$$

(c)

The period of state i is defined by

$$d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}.$$

Starting from state 2, the following return paths have positive probability:

$$2 \rightarrow 3 \rightarrow 2 \quad \text{and} \quad 2 \rightarrow 3 \rightarrow 3 \rightarrow 2.$$

Thus, returns are possible in both two and three steps. Since $\gcd(2, 3) = 1$,

$$d(2) = 1.$$

All three states communicate, so the chain is irreducible and all states have the same period. Therefore, the period of the whole chain is 1.

(d)

By the multiplication rule and the Markov property,

$$\begin{aligned} P(X_0 = 3, X_1 = 2, X_2 = 1) &= P(X_0 = 3) P(X_1 = 2 \mid X_0 = 3) P(X_2 = 1 \mid X_1 = 2) \\ &= \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} \\ &= \frac{1}{40}. \end{aligned}$$

(e)

Since there are two transitions from time 3 to time 5,

$$P(X_5 = j \mid X_3 = 1) = (P^2)_{1j} = \sum_{k=1}^3 p_{1k} p_{kj}.$$

Consequently,

$$P(X_5 = 1 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{16},$$

$$P(X_5 = 2 \mid X_3 = 1) = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16},$$

$$P(X_5 = 3 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} = \frac{6}{16}.$$

The conditional expectation is

$$\begin{aligned} E(X_5 \mid X_3 = 1) &= 1 \cdot \frac{7}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{6}{16} \\ &= \frac{31}{16}. \end{aligned}$$

The conditional second moment is

$$\begin{aligned} E(X_5^2 \mid X_3 = 1) &= 1^2 \cdot \frac{7}{16} + 2^2 \cdot \frac{3}{16} + 3^2 \cdot \frac{6}{16} \\ &= \frac{73}{16}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(X_5 \mid X_3 = 1) &= E(X_5^2 \mid X_3 = 1) - [E(X_5 \mid X_3 = 1)]^2 \\ &= \frac{73}{16} - \left(\frac{31}{16}\right)^2 \\ &= \frac{207}{256}. \end{aligned}$$

Problem 6. Estimating a coin's head probability

Assume $n \geq 1$. Given p , the number of heads satisfies

$$X \mid p \sim \text{Binomial}(n, p).$$

Hence, for the observation $X = k$,

$$P(X = k \mid p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq k \leq n.$$

(a) Maximum Likelihood Estimation (MLE)

Treating p as an unknown constant, the likelihood function is

$$L(p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq p \leq 1.$$

For $0 < k < n$, the log-likelihood is

$$\ell(p) = \log \binom{n}{k} + k \log p + (n-k) \log(1-p).$$

Differentiating and setting the derivative equal to zero gives

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0,$$

so

$$k(1-p) = (n-k)p \implies p = \frac{k}{n}.$$

Moreover,

$$\ell''(p) = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2} < 0,$$

so this point maximizes the likelihood.

If $k = 0$, the likelihood is maximized at $p = 0$; if $k = n$, it is maximized at $p = 1$. Thus, in all cases,

$$\hat{p}_{\text{MLE}} = \frac{k}{n}.$$

(b) Maximum A Posteriori Estimation (MAP)

The prior distribution is

$$p \sim \text{Beta}(1, 1),$$

which is the uniform distribution on $[0, 1]$. Therefore, the prior density is $f(p) = 1$ on this interval.

By Bayes' rule,

$$f(p \mid X = k) \propto P(X = k \mid p) f(p) \propto p^k (1-p)^{n-k}.$$

Consequently,

$$p \mid X = k \sim \text{Beta}(k+1, n-k+1).$$

The MAP estimate maximizes the posterior density. Since the prior density is constant, maximizing the posterior density is equivalent to maximizing the likelihood. Hence,

$$\hat{p}_{\text{MAP}} = \frac{k}{n},$$

including the boundary cases $k = 0$ and $k = n$.

(c) Minimum Mean Squared Error Estimation (MMSE)

The MMSE estimate is the posterior mean:

$$\hat{p}_{\text{MMSE}} = E[p \mid X = k].$$

Since

$$\text{Unif}(0, 1) = \text{Beta}(1, 1),$$

the posterior distribution is again

$$p \mid X = k \sim \text{Beta}(k + 1, n - k + 1).$$

The mean of a $\text{Beta}(\alpha, \beta)$ distribution is $\alpha/(\alpha + \beta)$. Therefore,

$$\hat{p}_{\text{MMSE}} = \frac{k + 1}{n + 2}.$$

(d) Linear Least Squares Estimation (LLSE)

Using the given formula with $Y = p$, we obtain

$$\hat{p}_{\text{LLSE}} = E[p] + \frac{\text{Cov}(X, p)}{\text{Var}(X)}(k - E[X]).$$

All moments below are computed under the joint distribution specified by the prior and the observation model.

Since $p \sim \text{Unif}(0, 1)$,

$$E[p] = \frac{1}{2}, \quad E[p^2] = \frac{1}{3}, \quad \text{Var}(p) = \frac{1}{12}.$$

Also, $E[X \mid p] = np$, so the law of total expectation gives

$$E[X] = E[E[X \mid p]] = nE[p] = \frac{n}{2}.$$

By conditioning on p ,

$$E[Xp] = E[pE[X \mid p]] = nE[p^2] = \frac{n}{3}.$$

Thus,

$$\text{Cov}(X, p) = E[Xp] - E[X]E[p] = \frac{n}{3} - \frac{n}{2} \cdot \frac{1}{2} = \frac{n}{12}.$$

By the law of total variance,

$$\begin{aligned} \text{Var}(X) &= E[\text{Var}(X \mid p)] + \text{Var}(E[X \mid p]) \\ &= E[np(1 - p)] + \text{Var}(np) \\ &= n \left(\frac{1}{2} - \frac{1}{3} \right) + n^2 \left(\frac{1}{12} \right) \\ &= \frac{n(n + 2)}{12}. \end{aligned}$$

Therefore,

$$\begin{aligned}\hat{p}_{\text{LLSE}} &= \frac{1}{2} + \frac{n/12}{n(n+2)/12} \left(k - \frac{n}{2}\right) \\ &= \frac{1}{2} + \frac{k - n/2}{n+2} \\ &= \frac{k+1}{n+2}.\end{aligned}$$

Problem 7. Gaussian MMSE and LLSE

Method 1

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

where we assume that $\text{Var}(X) > 0$. Define

$$Z = Y - \mu_Y - a(X - \mu_X).$$

Then

$$Y = \mu_Y + a(X - \mu_X) + Z.$$

First, the residual Z has mean zero:

$$\begin{aligned}E[Z] &= E[Y] - \mu_Y - a(E[X] - \mu_X) \\ &= 0.\end{aligned}$$

Moreover,

$$\begin{aligned}\text{Cov}(X, Z) &= E[(X - E[X])(Z - E[Z])] \\ &= E[(X - \mu_X)Z] \\ &= E[(X - \mu_X)((Y - \mu_Y) - a(X - \mu_X))] \\ &= E[(X - \mu_X)(Y - \mu_Y)] - aE[(X - \mu_X)^2] \\ &= \text{Cov}(X, Y) - a \text{Var}(X) \\ &= 0.\end{aligned}$$

Since X and Y are jointly Gaussian and Z is an affine function of X and Y , the variables X and Z are also jointly Gaussian. Thus, their zero covariance implies that they are independent. Consequently,

$$E[Z | X] = E[Z] = 0.$$

Taking conditional expectations gives

$$\begin{aligned}E[Y | X] &= E[\mu_Y + a(X - \mu_X) + Z | X] \\ &= \mu_Y + a(X - \mu_X) + E[Z | X] \\ &= \mu_Y + a(X - \mu_X).\end{aligned}$$

To verify the minimum mean squared error property, write

$$m(X) = \mu_Y + a(X - \mu_X).$$

Since $Y = m(X) + Z$, the mean squared error of $m(X)$ is

$$E[(Y - m(X))^2] = E[Z^2].$$

Any other square-integrable predictor can be written as

$$g(X) = m(X) + d(X).$$

Using $E[Z | X] = 0$, we obtain

$$E[Zd(X)] = E[d(X)E[Z | X]] = 0.$$

Therefore,

$$\begin{aligned} E[(Y - g(X))^2] &= E[(Z - d(X))^2] \\ &= E[Z^2] - 2E[Zd(X)] + E[d(X)^2] \\ &= E[Z^2] + E[d(X)^2] \\ &\geq E[Z^2]. \end{aligned}$$

Hence $m(X) = E[Y | X]$ is the MMSE predictor. Since $m(X)$ is also affine in X , it is the best linear least-squares predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X])$$

Method 2

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad \sigma_X^2 = \text{Var}(X) > 0, \quad \sigma_Y^2 = \text{Var}(Y).$$

First assume $\sigma_Y > 0$. Define

$$U = \frac{X - \mu_X}{\sigma_X}, \quad V = \frac{Y - \mu_Y}{\sigma_Y}, \quad \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

For $|\rho| < 1$, the joint density of U, V is

$$f_{U,V}(u, v) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)}\right).$$

Given $U = u$, treat u as a constant. Completing the square in v , we get

$$u^2 - 2\rho uv + v^2 = (v - \rho u)^2 + (1 - \rho^2)u^2.$$

Therefore,

$$f_{U,V}(u, v) = \frac{e^{-u^2/2}}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{(v - \rho u)^2}{2(1-\rho^2)}\right).$$

Since $f_{V|U}(v | u) = f_{U,V}(u, v)/f_U(u)$, all factors independent of v can be absorbed into a constant. Thus,

$$f_{V|U}(v | u) \propto \exp\left(-\frac{(v - \rho u)^2}{2(1 - \rho^2)}\right).$$

This is a Gaussian density symmetric about $v = \rho u$. Hence its mean is

$$E[V | U = u] = \rho u.$$

To return to the original variables, note that

$$Y = \mu_Y + \sigma_Y V.$$

Also, $X = x$ is equivalent to $U = (x - \mu_X)/\sigma_X$. Therefore,

$$\begin{aligned} E[Y | X = x] &= \mu_Y + \sigma_Y E[V | X = x] \\ &= \mu_Y + \sigma_Y E\left[V \mid U = \frac{x - \mu_X}{\sigma_X}\right] \\ &= \mu_Y + \sigma_Y \rho \frac{x - \mu_X}{\sigma_X} \\ &= \mu_Y + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(x - \mu_X). \end{aligned}$$

The conditional expectation minimizes mean squared error among all predictors based on X . Since the expression above is affine in X , it is also the best linear predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

Degenerate cases. If $\sigma_Y = 0$, then $Y = \mu_Y$ almost surely and the result is immediate. If $\sigma_X, \sigma_Y > 0$ and $|\rho| = 1$, then

$$\text{Var}(V - \rho U) = 1 - \rho^2 = 0,$$

so $V = \rho U$ almost surely, giving the same affine conditional mean. If $\text{Var}(X) = 0$, the displayed covariance ratio is undefined, but both the MMSE and LLSE predicted values are $E[Y]$.

Problem 8. Expected return of the uniform-sum program

Let $X_1, X_2, \dots$ be independent random variables, each uniformly distributed on $(0, 1)$. Define

$$S_n = \sum_{i=1}^n X_i, \quad N = \min\{n \geq 1 : S_n > 1\}.$$

Here, N is the number of loop iterations, and the program returns S_N .

For a positive integer-valued random variable N , the definition of expectation gives

$$\mathbb{E}[N] = \Pr(N = 1) + 2\Pr(N = 2) + 3\Pr(N = 3) + \dots$$

Splitting each coefficient into a sum of ones and regrouping the nonnegative terms, we obtain

$$\begin{aligned}
 \mathbb{E}[N] &= [\Pr(N = 1) + \Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 3) + \cdots] + \cdots \\
 &= \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \sum_{n=0}^{\infty} \Pr(N > n).
 \end{aligned}$$

Since the increments are nonnegative,

$$\Pr(N > n) = \Pr(S_n \leq 1) = \frac{1}{n!},$$

where, for $n \geq 1$, the last equality follows from the volume of the region

$$\left\{ (x_1, \dots, x_n) : x_i \geq 0, \sum_{i=1}^n x_i \leq 1 \right\}.$$

For $n = 0$, the probability is $1 = 1/0!$. Therefore,

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = e.$$

We can derive the expected return value directly, without invoking Wald's equation. Define the actual contribution of the i th iteration by

$$Y_i = \begin{cases} X_i, & \text{if the } i\text{th iteration occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$Y_i = X_i \mathbf{1}_{\{N \geq i\}}.$$

The key observation is that whether the i th iteration occurs depends only on $X_1, \dots, X_{i-1}$, not on X_i . Hence the event $\{N \geq i\}$ is independent of X_i .

Since

$$\mathbb{E}[X_i] = \frac{1}{2},$$

we have

$$\mathbb{E}[Y_i] = \mathbb{E}[X_i] \Pr(N \geq i) = \frac{1}{2} \Pr(N \geq i).$$

The returned sum is

$$S_N = \sum_{i=1}^{\infty} Y_i.$$

All the terms are nonnegative, so we may interchange expectation and summation:

$$\begin{aligned}
 \mathbb{E}[S_N] &= \sum_{i=1}^{\infty} \mathbb{E}[Y_i] \\
 &= \frac{1}{2} \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \frac{1}{2} \mathbb{E}[N] \\
 &= \frac{e}{2} \approx 1.35914.
 \end{aligned}$$

Problem 9. Hoeffding's lemma

Suppose that $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely. Then, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

If $a = b$, the zero-mean assumption implies that $X = 0$ almost surely, so the result is immediate. Henceforth, assume that $a < b$.

Since $\mathbb{E}[X] = 0$ and $a \leq X \leq b$, we have $a \leq 0 \leq b$. For every $x \in [a, b]$,

$$x = \frac{b-x}{b-a}a + \frac{x-a}{b-a}b.$$

The two coefficients are nonnegative and sum to one. Therefore, by the convexity of $x \mapsto e^{\lambda x}$,

$$e^{\lambda x} \leq \frac{b-x}{b-a}e^{\lambda a} + \frac{x-a}{b-a}e^{\lambda b}.$$

Substituting $x = X$, taking expectations, and using $\mathbb{E}[X] = 0$, we obtain

$$\mathbb{E}[e^{\lambda X}] \leq \frac{b}{b-a}e^{\lambda a} + \frac{-a}{b-a}e^{\lambda b}.$$

Let

$$p = \frac{-a}{b-a} \in [0, 1], \quad t = \lambda(b-a) > 0.$$

Then

$$1-p = \frac{b}{b-a}, \quad \lambda a = -pt, \quad \lambda b = (1-p)t.$$

Consequently,

$$\mathbb{E}[e^{\lambda X}] \leq (1-p)e^{-pt} + pe^{(1-p)t}.$$

Define

$$h(t) = \log\left((1-p)e^{-pt} + pe^{(1-p)t}\right) = -pt + \log(1-p + pe^t).$$

It suffices to prove that

$$h(t) \leq \frac{t^2}{8}.$$

First, observe that

$$h(0) = 0.$$

Differentiating once gives

$$h'(t) = -p + \frac{pe^t}{1 - p + pe^t},$$

and hence

$$h'(0) = 0.$$

Differentiating again gives

$$h''(t) = \frac{p(1-p)e^t}{(1-p+pe^t)^2}.$$

Let

$$q(t) = \frac{pe^t}{1-p+pe^t} \in [0, 1].$$

Then

$$h''(t) = q(t)(1 - q(t)).$$

For every $q \in [0, 1]$,

$$q(1 - q) = \frac{1}{4} - \left(q - \frac{1}{2}\right)^2 \leq \frac{1}{4}.$$

Therefore,

$$h''(t) \leq \frac{1}{4}.$$

Using $h'(0) = 0$, for every $s \geq 0$ we have

$$h'(s) = \int_0^s h''(u) du \leq \frac{s}{4}.$$

Using $h(0) = 0$ and integrating once more, we obtain

$$h(t) = \int_0^t h'(s) ds \leq \int_0^t \frac{s}{4} ds = \frac{t^2}{8}.$$

It follows that

$$\mathbb{E}[e^{\lambda X}] \leq e^{h(t)} \leq e^{t^2/8} = \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Problem 10. Utility, fairness, and RL rewards

Throughout the solution, the feasible set is

$$F = \{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : x_1 + 2x_2 \leq 12\},$$

and the designer evaluates an allocation through

$$W(x_1, x_2) = w_1 u_1(x_1) + w_2 u_2(x_2), \quad w_1, w_2 > 0.$$

All logarithms are natural, and logarithmic utilities are considered only at strictly positive service rates.

(a)

What a utility function represents. Write $a \succeq b$ when outcome a is at least as desirable as outcome b . A utility function u represents this preference relation if

$$a \succeq b \iff u(a) \geq u(b).$$

Thus, utility assigns numbers to outcomes in a way that reproduces their preference ordering. On the positive service rates, both $u(x) = x$ and $v(x) = \log x$ represent the preference that more service is better, since both functions are strictly increasing.

This does not mean that summing these utilities across users produces the same optimal allocation. With equal weights, consider

$$\begin{aligned} W_u(x_1, x_2) &= x_1 + x_2, \\ W_v(x_1, x_2) &= \log x_1 + \log x_2 = \log(x_1 x_2). \end{aligned}$$

On the given feasible set, the first objective is maximized at $(12, 0)$, whereas the logarithmic objective, on the positive part of the feasible set, is maximized at $(6, 3)$. These optima are derived explicitly in part (c). Hence, even though the individual deterministic preference orderings agree wherever both utilities are defined, the aggregate objectives need not give the same allocation.

Ordering outcomes versus measuring contributions. The statement “more service is better” specifies the direction of preference, but it does not specify how much additional utility a given increase in service should contribute. The distinction can be seen from the slopes:

$$u'(x) = 1, \quad v'(x) = \frac{1}{x}.$$

For a small increase $h > 0$, linear utility increases by exactly h , whereas logarithmic utility increases by approximately h/x . Under the logarithmic scale, the same service increment therefore contributes more when the user’s existing service rate is small.

When resources are limited, the designer must compare improvements to different users and decide which improvement contributes more to the aggregate objective. An individual’s ordering of their own outcomes does not supply this interpersonal comparison. The chosen utility scales and weights supply additional information about how users’ gains are to be traded off. In particular, a strictly increasing transformation preserves an individual’s ordinal preferences, but applying it to each user’s utility before summing can change the maximizer of the sum.

(b)

Increasing utility with diminishing slope. The condition $u'(x) > 0$ means that additional service always increases utility. The condition $u''(x) < 0$ means that the slope $u'(x)$ strictly decreases as service increases: utility continues to grow, but it grows more slowly. This is strict concavity, or diminishing marginal utility. A representative curve rises from left to right and becomes progressively flatter; its tangent is steeper at a smaller service rate than at a larger one.

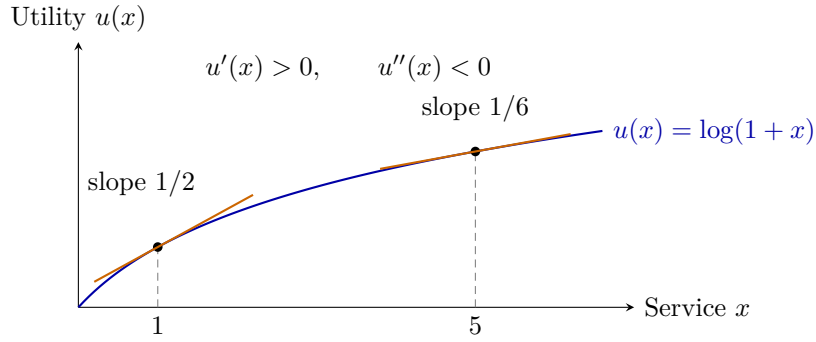


Figure 1: Increasing, strictly concave utility: the same extra service has a smaller marginal benefit at a higher service level.

Why a transfer increases the sum of utilities. Suppose $0 < a < b$, and transfer an amount δ from the better-served user to the worse-served user, where

$$0 < \delta < \frac{b-a}{2}.$$

Define the utility along this transfer by

$$h(t) = u(a+t) + u(b-t), \quad 0 \leq t \leq \frac{b-a}{2}.$$

Differentiating gives

$$h'(t) = u'(a+t) - u'(b-t).$$

For $0 \leq t < (b-a)/2$, we have $a+t < b-t$. Since u' is strictly decreasing, it follows that

$$u'(a+t) > u'(b-t), \quad h'(t) > 0.$$

Consequently,

$$\begin{aligned} & u(a+\delta) + u(b-\delta) - u(a) - u(b) \\ &= \int_0^\delta (u'(a+t) - u'(b-t)) dt > 0. \end{aligned}$$

The worse-served user gains more utility from each transferred unit than the better-served user loses. This proves the required strict increase.

Continuing the transfer until

$$\delta = \frac{b-a}{2}$$

gives equal service rates:

$$a + \delta = b - \delta = \frac{a+b}{2}.$$

For a fixed total amount of service $a+b$, this is the maximum of the sum of the two identical strictly concave utilities. Equivalently, strict concavity gives

$$u(a) + u(b) < 2u\left(\frac{a+b}{2}\right) \quad \text{when } a \neq b.$$

Equal service is not the same as equal resource use. The preceding argument holds the sum of the service rates fixed. It does not account for a difference in the resources needed to deliver those rates. If both users cost the same per unit of service, a one-for-one service transfer preserves resource consumption. In this problem, however, a unit of service to user 2 costs twice as much as a unit to user 1.

For example, at the logarithmic optimum $(6, 3)$, transferring $\delta > 0$ units of service from user 1 to user 2 would give $(6 - \delta, 3 + \delta)$. Its resource consumption is

$$(6 - \delta) + 2(3 + \delta) = 12 + \delta > 12,$$

so the transfer is infeasible. A resource-preserving increase of one unit in user 2's service instead requires a decrease of two units in user 1's service. Thus the fixed-service-sum argument cannot simply be applied to the resource boundary.

Indeed, $u(x) = \log x$ is strictly increasing and strictly concave, but its sum is maximized at $(6, 3)$, not at equal service rates. The equal-service allocation on the resource boundary is $(4, 4)$, and

$$\log 6 + \log 3 = \log 18 > \log 16 = 2 \log 4.$$

Strict concavity favors balancing marginal gains subject to feasibility; it does not, by itself, require equal service rates under unequal costs.

(c)

The three optimal allocations. For total service, the constraint implies

$$x_1 + x_2 = (x_1 + 2x_2) - x_2 \leq 12 - x_2 \leq 12.$$

Equality requires $x_2 = 0$ and $x_1 = 12$. Therefore, the unique maximizer of total service is

$$(x_1, x_2) = (12, 0).$$

The objective places all resources with the user who is cheaper to serve.

For logarithmic utility, maximizing $\log x_1 + \log x_2$ is equivalent to maximizing $x_1 x_2$ over positive feasible rates. By the arithmetic–geometric mean inequality,

$$x_1 x_2 = \frac{x_1(2x_2)}{2} \leq \frac{1}{2} \left(\frac{x_1 + 2x_2}{2} \right)^2 \leq 18.$$

Equality holds precisely when $x_1 = 2x_2$ and $x_1 + 2x_2 = 12$. Hence,

$$(x_1, x_2) = (6, 3)$$

is the unique logarithmic-utility maximizer.

For minimum service, put $m = \min\{x_1, x_2\}$. Both rates are at least m , so

$$3m \leq x_1 + 2x_2 \leq 12, \quad m \leq 4.$$

The bound is attained at $(4, 4)$. Moreover, attaining $m = 4$ requires both rates to be at least 4, and the resource constraint then forces both to equal 4. The unique optimum is therefore

$$(x_1, x_2) = (4, 4).$$

The first objective maximizes total service, even at the cost of giving one user no service. The third maximizes the worst-served user's rate and, in this particular feasible set, produces equal service. The logarithmic objective produces neither of those allocations. Its principle is *proportional fairness*, rather than maximum total service or absolute equality of service.

The proportional-fairness inequality. Let $C \subseteq \mathbb{R}_{\geq 0}^n$ be convex, and suppose that $\mathbf{x}^*$ maximizes

$$W(\mathbf{x}) = \sum_{i=1}^n \log x_i$$

over C . Fix any $\mathbf{y} \in C$. Convexity allows us to move along the line segment from $\mathbf{x}^*$ to $\mathbf{y}$ while staying feasible:

$$\mathbf{x}(t) = \mathbf{x}^* + t(\mathbf{y} - \mathbf{x}^*) \in C, \quad 0 \leq t \leq 1.$$

Define

$$\phi(t) = W(\mathbf{x}(t)) = \sum_{i=1}^n \log(x_i^* + t(y_i - x_i^*)).$$

Optimality implies $\phi(t) \leq \phi(0)$ for every $t \in [0, 1]$. Thus,

$$\phi'_+(0) = \lim_{t \downarrow 0} \frac{\phi(t) - \phi(0)}{t} \leq 0.$$

Computing this directional derivative yields

$$\sum_{i=1}^n \frac{y_i - x_i^*}{x_i^*} \leq 0.$$

Each summand is user i 's proportional change in service when moving from $\mathbf{x}^*$ to $\mathbf{y}$. The inequality says that no feasible alternative can make the sum of these proportional changes positive: proportional gains must be offset by proportional losses. It does not say that every individual user must lose. This is the proportional-fairness criterion.

Weighted logarithmic utility. For $W(x_1, x_2) = 2 \log x_1 + \log x_2$, maximizing utility is equivalent to maximizing $x_1^2 x_2$. For the arithmetic-geometric mean inequality, split x_1 into two equal terms:

$$\begin{aligned} x_1^2 x_2 &= 2 \left(\frac{x_1}{2} \right) \left(\frac{x_1}{2} \right) (2x_2) \\ &\leq 2 \left(\frac{x_1/2 + x_1/2 + 2x_2}{3} \right)^3 \\ &= 2 \left(\frac{x_1 + 2x_2}{3} \right)^3 \leq 2 \left(\frac{12}{3} \right)^3 = 128. \end{aligned}$$

Equality requires

$$\frac{x_1}{2} = \frac{x_1}{2} = 2x_2, \quad x_1 + 2x_2 = 12.$$

Therefore, $x_1 = 4x_2$, giving the unique optimum

$$(x_1, x_2) = (8, 2).$$

The coefficient 2 gives user 1's utility twice the relative importance in the aggregate objective. Algebraically, it is as if that utility were counted twice. It need not mean that there are literally two physical users of type 1; such an interpretation would require an additional modeling assumption. The weight specifies the designer's trade-off, not the service cost, which is already encoded in the constraint.

(d)

Expected payoffs and expected logarithmic utilities. Let $X_A = 5$ with certainty, while X_B takes values 1 and 9 with equal probabilities. Their expected payoffs are

$$\mathbb{E}[X_A] = 5, \quad \mathbb{E}[X_B] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 = 5.$$

Thus, maximizing expected payoff gives indifference between the options. Their expected logarithmic utilities are instead

$$\begin{aligned} \mathbb{E}[\log X_A] &= \log 5, \\ \mathbb{E}[\log X_B] &= \frac{1}{2} \log 1 + \frac{1}{2} \log 9 = \log 3. \end{aligned}$$

Since $\log 5 > \log 3$, expected logarithmic utility strictly prefers the certain payoff, Option A.

The logarithmic certainty equivalent of Option B is the certain amount c whose utility equals the lottery's expected utility:

$$\log c = \mathbb{E}[\log X_B] = \log 3, \quad c = 3.$$

Thus, the lottery is equivalent to a certain payoff of 3 under logarithmic utility, even though its expected payoff is 5.

Jensen's inequality and risk aversion. For a concave function u , Jensen's inequality states, whenever the quantities are well defined and finite, that

$$\mathbb{E}[u(X)] \leq u(\mathbb{E}[X]).$$

The right-hand side is the utility of receiving the expected payoff with certainty. Hence an expected-utility decision maker with concave utility weakly prefers that certain payoff to the lottery. For strictly concave utility and a nonconstant lottery, the inequality is strict.

If u is also strictly increasing and the certainty equivalent satisfies $u(c) = \mathbb{E}[u(X)]$, then

$$u(c) \leq u(\mathbb{E}[X]) \implies c \leq \mathbb{E}[X].$$

For logarithmic utility this becomes

$$c = \exp(\mathbb{E}[\log X]) \leq \mathbb{E}[X].$$

This is the risk-aversion interpretation of concavity: uncertainty lowers the certainty equivalent relative to the expected payoff.

Why this is different from interpersonal fairness. Here there is only one user, and utility is averaged over that user's possible outcomes in different states of uncertainty. In part (b), utilities are added across different users receiving a deterministic allocation. The same diminishing-slope property explains both results, but the comparisons are different: risk aversion concerns uncertainty for one person, while fairness concerns the distribution across people. There is no interpersonal allocation to judge as fair or unfair in the single-user lottery comparison.

(e)

Lotteries and the four preference axioms. Let the finite outcome set be $\mathcal{X} = \{z_1, \dots, z_n\}$. A lottery is a probability vector

$$p = (p_1, \dots, p_n), \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1,$$

where p_i is the probability of outcome z_i . The standard axioms on preferences over all such lotteries are the following.

1. **Completeness.** For any lotteries p, q , either $p \succeq q$ or $q \succeq p$, or both. Thus every pair of lotteries can be compared; when both relations hold, write $p \sim q$.
2. **Transitivity.** If $p \succeq q$ and $q \succeq r$, then $p \succeq r$. The preference ordering is internally consistent.
3. **Continuity.** If $p \succ q \succ r$, there exists $\alpha \in (0, 1)$ such that

$$q \sim \alpha p + (1 - \alpha)r.$$

An intermediate lottery is indifferent to some probability mixture of the better and worse lotteries; preferences do not have an unbridgeable jump between them.

4. **Independence.** For any lotteries p, q, r and $\alpha \in (0, 1)$,

$$p \succeq q \iff \alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r.$$

Mixing each of two lotteries with the same third lottery in the same proportion does not change their ordering.

Expected-utility representation. The von Neumann–Morgenstern theorem states that these axioms characterize preferences that admit an expected-utility representation. There exists $u : \mathcal{X} \rightarrow \mathbb{R}$ such that, writing

$$U(p) = \mathbb{E}_p[u(Z)] = \sum_{i=1}^n p_i u(z_i),$$

we have

$$p \succeq q \iff U(p) \geq U(q).$$

The decision maker can therefore be represented as choosing a lottery with maximal expected utility. Conversely, preferences represented in this way satisfy the four axioms.

Uniqueness up to positive affine transformations. If u represents these lottery preferences, then so does

$$v(z) = au(z) + b, \quad a > 0,$$

since

$$\mathbb{E}_p[v(Z)] = a\mathbb{E}_p[u(Z)] + b.$$

The same positive scaling and constant shift apply to every lottery, so no comparison changes. Moreover, any other expected-utility representation of the same preferences on all lotteries over $\mathcal{X}$ is related to u by such a positive affine transformation.

The theorem does not require u to be concave, and does not require it to be logarithmic. Concavity is an additional property relevant to risk attitudes when outcomes have a numerical payoff interpretation. Nor does the theorem determine a fair allocation across several users: it represents a decision maker's lottery preferences, without selecting the interpersonal utility scales, weights, or fairness objective of a system designer.

Why an arbitrary increasing transformation is not enough. For deterministic outcomes, a strictly increasing transformation preserves the ordering of utility values. In particular, $u(x) = x$ and $v(x) = \log x$ both give

$$9 \succ 5 \succ 1.$$

However, expected utility averages the transformed values over possible outcomes, and a nonlinear transformation can change comparisons between these averages.

The lotteries in part (d) show this directly. With $u(x) = x$,

$$\mathbb{E}[u(X_A)] = 5 = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 = \mathbb{E}[u(X_B)],$$

so $A \sim B$. With $v(x) = \log x$,

$$\mathbb{E}[v(X_A)] = \log 5 > \log 3 = \mathbb{E}[v(X_B)],$$

so $A \succ B$. Thus the deterministic rankings coincide, but the lottery rankings do not. This is why VNM utility is unique up to positive affine transformations, rather than arbitrary strictly increasing ones.

(f)

The reward hypothesis and expected utility. The reward hypothesis says that an agent's goal can be expressed as the maximization of the expected cumulative sum of a scalar reward signal. For this undiscounted, finite-horizon problem, the objective is

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

The modeling task is to choose rewards that quantify the intended goal, so that maximizing this expectation corresponds to pursuing that goal.

This resembles the VNM representation: an abstract preference is represented numerically, and choices are evaluated by an expectation. VNM uses the expected utility of

an outcome; in sequential decision making, an entire trajectory can be treated as the outcome, and its cumulative reward can serve as its utility. The VNM conclusion is conditional on its preference axioms, whereas the reward hypothesis is a claim about representing goals in the reward framework.

An expected-utility representation does not automatically give a stationary Markov reward on a prescribed state space. Such a reward has a fixed form $r(s, a, s')$, depending only on the current transition, with no additional dependence on the past or on an external time index. A utility assigned to a whole trajectory may depend on history that the specified state does not retain. Consequently, a trajectory-level utility need not decompose into stationary Markov rewards on that original state space. Additional state information may be necessary.

Fairness at each time versus fairness of average service. The two objectives are

$$J_{\text{instant}}(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} (\log X_1(t) + \log X_2(t)) \right]$$

and

$$J_{\text{average}}(\pi) = \mathbb{E}_{\pi} [\log \bar{X}_1 + \log \bar{X}_2],$$

$$\bar{X}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_i(t).$$

The first applies the utility function at every time step and then sums. The second first averages each user's service over time and only then applies utility. Since the logarithm is nonlinear, these operations cannot generally be interchanged.

For $T = 2$, all allocations in the two given schedules are feasible and use the full resource budget:

$$6 + 2 \cdot 3 = 12, \quad 10 + 2 \cdot 1 = 12, \quad 2 + 2 \cdot 5 = 12.$$

The schedules are deterministic, so the expectations do not change the following calculations.

For π_A , the allocation is $(6, 3)$ at both time steps. Therefore,

$$\begin{aligned} J_{\text{instant}}(\pi_A) &= 2(\log 6 + \log 3) \\ &= 2 \log 18 = \log 324. \end{aligned}$$

Its average service rates are

$$\bar{X}_1 = \frac{6+6}{2} = 6, \quad \bar{X}_2 = \frac{3+3}{2} = 3,$$

so

$$J_{\text{average}}(\pi_A) = \log 6 + \log 3 = \log 18.$$

For π_B , the allocations are $(10, 1)$ and $(2, 5)$. Thus,

$$\begin{aligned} J_{\text{instant}}(\pi_B) &= \log 10 + \log 1 + \log 2 + \log 5 \\ &= \log 100. \end{aligned}$$

However, its average service rates are also

$$\bar{X}_1 = \frac{10+2}{2} = 6, \quad \bar{X}_2 = \frac{1+5}{2} = 3.$$

Consequently,

$$J_{\text{average}}(\pi_B) = \log 6 + \log 3 = \log 18.$$

The instantaneous objective strictly prefers π_A , because $\log 324 > \log 100$, while the average-service objective is indifferent between the schedules. They are therefore not equivalent even as preference criteria: one gives a strict preference where the other gives a tie. Under the average-service objective, a user's low service at one time can be compensated by high service at another time.

A reward that exactly implements average-service utility. A direct construction is to give zero reward before the terminal transition and put the utility of the final averages in the terminal reward:

$$R_{t+1} = 0 \quad (0 \leq t < T-1), \quad R_T = \log \bar{X}_1 + \log \bar{X}_2.$$

For every realized trajectory,

$$\sum_{t=0}^{T-1} R_{t+1} = \log \bar{X}_1 + \log \bar{X}_2.$$

Taking the policy-dependent expectation gives exactly

$$\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right] = J_{\text{average}}(\pi).$$

Thus the construction works for stochastic policies and trajectories as well as for the two deterministic examples.

What the state must remember. The terminal reward depends on allocations throughout the episode. Rather than the whole history, it is enough to retain the cumulative service totals

$$C_i(t) = \sum_{\tau=0}^{t-1} X_i(\tau), \quad C_i(0) = 0, \quad i = 1, 2.$$

They update recursively as

$$C_i(t+1) = C_i(t) + X_i(t),$$

and the terminal reward can be written as

$$R_T = \log \left(\frac{C_1(T)}{T} \right) + \log \left(\frac{C_2(T)}{T} \right).$$

If the original state is S_t , augment it to

$$\tilde{S}_t = (S_t, C_1(t), C_2(t), t).$$

The cumulative totals retain exactly the past information required by this objective, and the time coordinate determines when the terminal reward is due. On a transition ending at time T , the reward reads the new totals; on every other transition it is zero. This is a fixed rule on augmented transitions, rather than a rule requiring access to an unrecorded history.

Provided that S_t already suffices for the Markov transition dynamics and the allocation mechanism, this augmentation makes the reward Markov. Including time also allows the finite-horizon reward rule to be expressed as a stationary function on the augmented state space. The need for these extra variables illustrates why an expected-utility representation alone does not supply a stationary Markov reward on an arbitrary original state space.

(g)

VNM expected utility represents preferences over uncertain outcomes by maximizing expected utility, under the four stated axioms. The reward hypothesis applies a related representation idea to sequential decisions: a goal is expressed through expected cumulative scalar reward. A policy induces a distribution over complete trajectories, so a trajectory can play the role of the uncertain outcome.

For example, if a trajectory τ is evaluated by

$$U(\tau) = \log \bar{X}_1(\tau) + \log \bar{X}_2(\tau),$$

then the terminal-reward construction in part (f) gives

$$\mathbb{E}_\pi[U(\tau)] = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

This makes the connection explicit, but it does not mean that VNM preferences automatically yield local, stationary Markov rewards. Trajectory utility may require relevant history to be included in the state, as the cumulative-service variables do here. Neither the theorem nor the reward hypothesis, by itself, selects which fairness objective the designer should adopt.